



Oligopoly Theory: An Introduction



Outline

- A Brief Step Back: Consumer Choice and Monopoly Theory
- The Foundational Models of Oligopoly Theory:
 - Bertrand Oligopoly
 - Cournot Oligopoly
- A Glimpse Towards Subsequent Models

Consumer Choice and Monopoly Theory: Premises

- Consumer choice is the branch of microeconomics analyzing how individuals maximize their satisfaction by making certain consumption decisions.
- Monopoly theory investigates instead the outcome of markets in which one firm is the only supplier of a particular good.

Consumer Choice: Utility Function

- A utility function is a mathematical rule reflecting consumer's tastes relating the quantity of goods purchased to the level of satisfaction experienced upon consumption.
- For instance, a generic individual $i = 1, \dots, n$ might have a utility function

$$u_i = \frac{\alpha_i q_i}{\beta_i} - \frac{q_i^2}{2\beta_i} + s_i \quad (1)$$

where q_i is the quantity consumed, p is the price of the good being purchased and s_i are the savings of the individual such that $s_i = w_i - pq_i$ with w_i being its wealth. α_i and β_i are just numbers depending on the consumer's personal tastes.

Consumer Choice: Demand Function I

- In a market with a generic number n of individuals, consuming only one kind of good purchased from a unique seller (monopolist), the demand function is derived from (1) as follows:
 1. The individual wants to choose the quantity of good to consume in such a way that its satisfaction - its utility function - reaches the maximum level, in other words, he wishes to solve a maximization problem

$$\max_{q_i} u_i$$

2. To do so it adopts a so-called First Order Condition, which is the derivative of (1) with respect to q_i equating zero

$$\frac{\alpha_i}{\beta_i} - \frac{q_i}{\beta_i} - p = 0$$

Consumer Choice: Demand Function II

3. To eventually isolate q_i

$$q_i = \alpha_i - \beta_i p \quad (2)$$

Equation (2) is the quantity maximizing the satisfaction of the individual, therefore it is the demand of i for the good sold by the monopolist, positive for $p < \frac{\alpha_i}{\beta_i}$.

4. Finally q_i is summed across all individuals, from $i = 1$ to $i = n$ to get the total demand of goods Q received by the monopolist:

$$Q = \sum_{i=1}^n q_i = \sum_{i=1}^n \alpha_i - \sum_{i=1}^n \beta_i p = a - bp \quad (3)$$



Monopoly Theory I

- A profit function is another mathematical rule reflecting the firm's structure relating the price of the good produced to the wealth generated upon sale.
- For instance, a generic monopolist might have a profit function

$$\pi = (p - c) Q \quad (4)$$

where c is the constant marginal cost of production faced by the monopolist.

- The monopoly price p^* imposed to consumers is then derived from (4) as follows:
 1. The monopolist wants to choose the price of its good in such a way that its profit is maximized, in other words he wishes to solve a maximization problem

$$\max_p \pi$$

Monopoly Theory II

2. To do so it adopts a FOC as well

$$a - bp^* + (p^* - c)(-b) = 0$$

3. To then isolate p^*

$$p^* = \frac{a}{2b} + \frac{c}{2} \quad (5)$$

Equation (5) is the price maximizing the profit function of the monopolist, therefore it is the price that will be imposed to consumers.

Monopoly Theory III

- Eventually the quantity of good sold by the monopolist will be obtained by plugging (5) into (3)

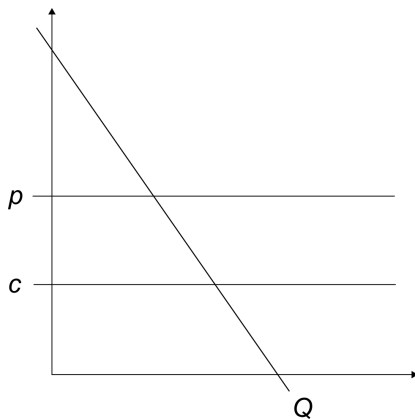
$$Q^* = \frac{a}{2} - \frac{bc}{2} \quad (6)$$

and the associated profit by plugging (5) and (6) into (4)

$$\pi^* = \frac{a^2}{4b} - \frac{ac}{2} + \frac{c^2b}{4} \quad (7)$$

with $\frac{a}{b} > c$ to ensure (6) and (7) positive.

Monopoly Theory: Graphical Intuition





Oligopoly Models: An Overview

- An oligopoly is a market featuring a small amount of firms, mutually interdependent, competing against each other.
- In a similar setting firms often end up colluding in order to maximize their profits.
- However, with fierce competition or with a proper exogenous intervention, oligopolies may still exist without turning into cartels.
- Oligopolies can either be *perfect*, featuring homogeneous goods, or *imperfect*, featuring differentiated goods.

Cournot Oligopoly: Premise

- A.A. Cournot, french mathematician and philosopher, in 1838 described a market in which firms compete on the amount of output produced, which they set simultaneously expecting such decision to have no impact on all other companies.
- Specifically, profit functions are defined for each firm, and then through partial differentiation a function representing a firm's best response for given (exogenous) output levels of the other firms in the market is built.
- Then, the BR functions constitute a system of linear equations which once it is solved yields a stable equilibrium solution - graphically such equilibrium occurs where these functions intersect.

Cournot Oligopoly: Profit Functions

- In a market with 2 identical firms, endowed with a constant marginal cost, the profit functions might be

$$\pi_1 = (p - c) q_1 \quad (8)$$

$$\pi_2 = (p - c) q_2 \quad (9)$$

with

$$p = \frac{a}{b} - \frac{1}{b} Q \quad (10)$$

from (3) where $Q = q_1 + q_2$.

Cournot Oligopoly: Best Reply Functions and Equilibrium Quantities I

- In a similar setting the equilibrium quantities for each firm q_1^* and q_2^* are found as follows:
 1. Firm 1 set the quantity maximizing the profit expecting firm 2 to maintain its quantity unchanged, in other words he wishes to solve a maximization problem

$$\max_{q_1} \pi_1$$

2. To do so it solves a FOC conditional on the best reply quantity set by the other firm q_2^{BR} , which means to differentiate its profit function with respect to q_1 treating q_2 as fixed to q_2^{BR} and equating to zero

$$-\frac{1}{b}q_1^{BR} + \frac{a}{b} - \frac{1}{b}q_1^{BR} - \frac{1}{b}q_2^{BR} - c = 0$$

Cournot Oligopoly: Best Reply Functions and Equilibrium Quantities II

3. Eventually, collecting q_1^{BR}

$$q_1^{BR} = \frac{a - bc}{2} - \frac{1}{2}q_2^{BR} \quad (11)$$

Equation (9) represents therefore the quantity set by firm 1 to maximize its profit expecting firm 2 to set q_2^{BR} and it is called firm 1 Best Reply function.

4. Firm 2 solves above steps symmetrically, yielding its Best Reply function

$$q_2^{BR} = \frac{a - bc}{2} - \frac{1}{2}q_1^{BR} \quad (12)$$

Cournot Oligopoly: Best Reply Functions and Equilibrium Quantities III

5. Once the Best Reply functions have been found, the equilibrium quantities are found at their intersection, or algebraically as the solution of the system

$$\begin{cases} q_1^{BR} = \frac{a-bc}{2} - \frac{1}{2}q_2^{BR} \\ q_2^{BR} = \frac{a-bc}{2} - \frac{1}{2}q_1^{BR} \end{cases}$$

which by backward substitution, among several other techniques, yields

$$q_1^* = q_2^* = \frac{a - bc}{3} \quad (13)$$

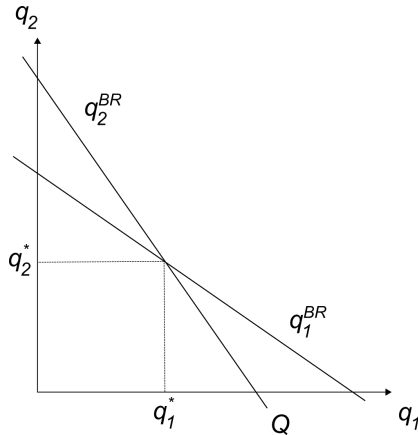
hence

$$Q^* = q_1^* + q_2^* = \frac{2(a - bc)}{3} \quad (14)$$

Cournot Oligopoly: Best Reply Functions and Equilibrium Quantities IV

- It is worth noting that this approach to the solution of a simultaneous decision-making process between firms might be interpreted in terms of Nash-Equilibria.

Cournot Oligopoly: Graphical Intuition



Cournot Oligopoly: Equilibrium Prices and Profits

- The analysis is concluded deriving the price and profits common to both firms. By substituting (14) in (10)

$$p^* = \frac{a}{3b} + \frac{2c}{3} \quad (15)$$

and by plugging (15) and (13) into (8) and into (9)

$$\pi_1^* = \frac{2(a - bc)^2}{9b} \quad (16)$$

$$\pi_2^* = \frac{2(a - bc)^2}{9b} \quad (17)$$

with $\frac{a}{b} > c$ to ensure (13), (14), (16) and (17) positive.

- It is worth noting that $p^* > c$.

Bertrand Oligopoly: Premise I

- The main result of Cournot model is that a single equilibrium price exists for all firms in the market, which is a consequence of the initial assumption of product homogeneity.
- Notwithstanding that duopolists set their quantities and not their prices, Cournot later writes that a firm might adjust its output by modifying its price, attracting more consumers, up to the point of cutting out the other firm.

Bertrand Oligopoly: Premise II

- But since in equilibrium only one price exists, above claim cast confusion onto the economic interpretation of its model and exposed himself to the "*Bertrand's Critique*" stating that firms would not reach a stable equilibrium as the price cut would go on indefinitely.
- Its 1883's critique was later formalized by Francis Y. Edgeworth who built a model where a firm sets its optimal price expecting the other's kept fixed to eventually reduce it down to marginal cost, achieving a stable equilibrium.



Bertrand Oligopoly: Premise III

- The reason why above model is now named after Joseph Bertrand is most likely due to a Irving Fisher review which in discussing similar models placed him first among Pareto, Marshall and Edgeworth himself (Dimand and Dore 1999).

Bertrand Oligopoly: Profit Functions

- In a market with 2 identical firms, endowed with a constant marginal cost, the profit functions might be

$$\pi_1 = (p_1 - c) q_1$$

$$\pi_2 = (p_2 - c) q_2$$

with

$$q_1 = \begin{cases} Q & \text{if } p_1 < p_2 \\ \frac{Q}{2} & \text{if } p_1 = p_2 \\ 0 & \text{if } p_1 > p_2 \end{cases}$$

$$q_2 = \begin{cases} Q & \text{if } p_2 < p_1 \\ \frac{Q}{2} & \text{if } p_2 = p_1 \\ 0 & \text{if } p_2 > p_1 \end{cases}$$

where $Q = q_1 + q_2$.

Bertrand Oligopoly: Equilibrium Prices I

- In such framework the equilibrium prices for each firm p_1^* and p_2^* might be found as in Cournot through BR functions intersection.
- Being the demand function not continuous, finding equilibrium algebraically is somewhat cumbersome, however the main intuition is that for any p_1^{BR} there exists a profitable deviation for firm 2, such that $p_2^{BR} = p_1^{BR} - \epsilon$, where ϵ is a very small number, until $p_2^{BR} = c = p_2^*$ and its profit becomes zero. The same is true for firm 1.

Therefore:

$$p_1^* = p_2^* = c \quad (18)$$

and

$$\pi_1^* = \pi_2^* = c \quad (19)$$

Bertrand Oligopoly: Graphical Intuition

It is worth noting that this approach to the solution of a simultaneous decision-making process between firms might be interpreted in terms of Nash-Equilibria. The outcome is that of a so-called *Prisoner's Dilemma Game* with the Nash Equilibrium $\neq$ Pareto Optimum.

	<i>Cooperate</i>	<i>Defect</i>
<i>Cooperate</i>	<i>Pareto Optimum</i> $p_1^* = p_2^* > c$	
<i>Defect</i>		<i>Nash Equilibrium</i> $p_1^* = p_2^* = c$

Following-Up

Several extensions to Pure Cournot and Bertrand Oligopoly have been proposed:

- Cournot and Bertrand with differentiated products;
- Bertrand in sequential games;
- Stackelberg;
- Spatial models:
 - Hotelling;
 - Salop;
 - Spokes;
 - Networks.

References

Dimand Robert and Mohammed Dore, 1999. "Cournot, Bertrand, and game theory: A Further Note". *Atlantic Economic Journal* 27 (3): 325-33.